

Operation Research

Linear Programming Problem (LPP)

- It is an optimization method applicable for solution of optimization problem where objective function and the constraints are linear.
- It was first applied in 1930 by economist, mainly solving resource allocation problem during World War II.

A Linear Programming Problem is one that concerned with finding the optimal value (maximum or minimum value) of a linear function (called objective function) of several variables (say x and y), subject to the conditions that the variables are non-negative and satisfy a set of linear inequalities (called linear constraints). The term linear implies that all the mathematical relations used in the problem are linear relations.

while the term programming refers to the method of determining a particular programme or plan of action.

Example
Eg. Production allocation problem (Mathematical Formulation)

- Manufacturer produces models M_1 and M_2
 - M_1 requires 4 hours of grinding and 2 hours polishing
 - M_2 requires 2 hours of grinding and 5 hours polishing
 - Manufacturer has 2 grinders and 3 polishers.
 - Each grinder works for 40 hours a week and each polisher works 60 hours a week
 - profit on an M_1 model is Rs 3.00 and on an M_2 is Rs 4.00 whatever produced in a week sold in market.
- Q. How should the manufacturer allocate his production capacity to the 2 type, of models so that he may make the maximum profit in a week.

$$Z = 3x_1 + 4x_2$$

$$4x_1 + 2x_2 \leq 80$$

$$2x_1 + 5x_2 \leq 180$$

$$x_1 \geq 0, x_2 \geq 0$$

Let x_1 be the no. of M_1 models and x_2 be the no. of M_2 models that the manufacturer decides to produce per week. Then his weekly profit (in rupees) is given by $Z = 3x_1 + 4x_2$

The total no. of grinding hours needed per week is given by, $4x_1 + 2x_2$.

The total no. of polishing hours needed per week is given by, $2x_1 + 5x_2$.

Since he does not have more than 80 hours of grinding nor does he have 180 hours of polishing.

Since no negatives $x_1 \geq 0$ and $x_2 \geq 0$

Mathematical form of production allocation problem is Maximize,

$$Z = 3x_1 + 4x_2 \text{ (objective)}$$

$$4x_1 + 2x_2 \leq 80$$

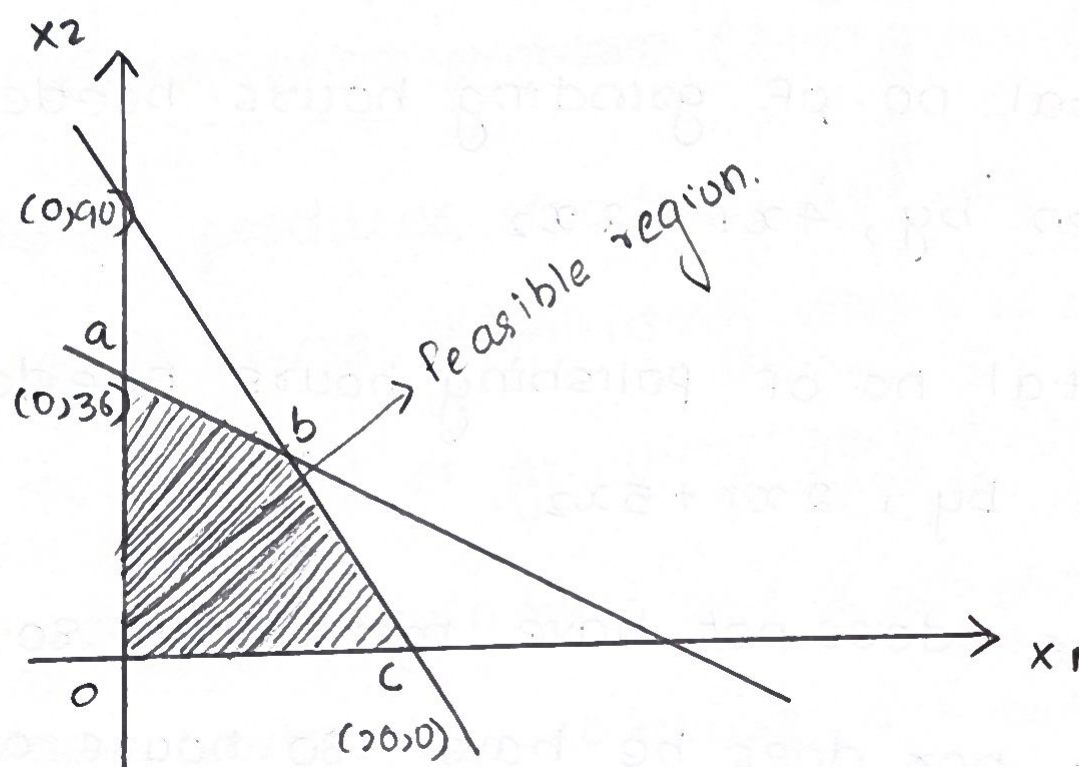
$$2x_1 + 5x_2 \leq 180$$

$$x_1 \geq 0, x_2 \geq 0$$

constraints.

Graphical Solution Method

We have to find (x_1, x_2) which satisfies the given constraints.



convert every inequalities to an equation

$$4x_1 + 2x_2 = 80 \text{ ——— ①}$$

$$3x_1 + 5x_2 = 180 \text{ ——— ②}$$

$$x_1 = 0, x_2 = 0$$

$$4x_1 + 2x_2 = 80$$

x_1	0	20
x_2	40	0

$$3x_1 + 5x_2 = 180$$

x_1	0	36
x_2	90	0

Feasible region is OABC

vertices of feasible region

(x_1, x_2)	z
$(0, 0)$	0
$(20, 0)$	60
$(0, 36)$	144
$(2.5, 35)$	147.5

$$4x_1 + 2x_2 = 80 \text{ --- ①}$$

$$\text{②} \times 2$$

$$4x_1 + 10x_2 = 360 \text{ --- ③}$$

$$\text{③} - \text{①}$$

$$8x_2 = 280$$

$$x_2 = \underline{\underline{35}}$$

$$x_1 = \frac{180-175}{2} = \frac{5}{2} = \underline{\underline{2.5}}$$

$\therefore$ solution is $x_1 = 2.5, x_2 = 35$ and

$$\underline{\underline{\max z = 147.5}}$$

General Linear Programming Problem

Let z be a linear function defined on $\mathbb{R}^n$ defined by

$$z = c_1x_1 + c_2x_2 + \dots + c_nx_n \quad \text{--- (a)}$$

where c_j 's are constraints.

Let (a_{ij}) be an $m \times n$ real matrix and let $\{b_1, b_2, \dots, b_m\}$ be a set of constraints such that

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \geq b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n \geq b_2$$

$\vdots$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n \geq b_m \quad \text{--- (b)}$$

$$\text{and let } x_j \geq 0, j = 1, 2, \dots, n \quad \text{--- (c)}$$

The problem of determining n tuple $(x_1, x_2, \dots, x_n)$ which makes z a minimum (or maximum) and which satisfies (b) and (c) is called general LPP.

o

The linear fn $z = c_1x_1 + c_2x_2 + \dots + c_nx_n$ which is minimized or maximized is called

objective function.

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$$

$$\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$$

- The inequality (b) are called constraints of the general LPP.
- An n tuple $(x_1, x_2, \dots, x_n)$ are called non-negative restrictions.
- An n tuple $(x_1, x_2, \dots, x_n)$ of real numbers which satisfies the constraints of a general LPP is called solution.
- Any solution to a general LPP which also satisfies the non negative constraint is called a feasible solution to the general LPP.
- Any feasible solution of a linear programming problem that ^{gives} maximum or minimum value of the objective function is called an optimal solution.

Remark

It is not necessary that the constraints in definition may be the inequations of the same type. Using the summation convention, in such a general situation, we may restate the constraints of the general L.P.P as follows.

$$1. \sum_{j=1}^n a_{ij} x_j \leq b_i \quad i = 1, 2, \dots, K$$

2. $\sum_{j=1}^n a_{ij} x_j \geq b_i \quad i = k+1, k+2, \dots, l$
3. $\sum_{j=1}^n a_{ij} x_j = b_i \quad i = 1, 1+1, \dots, m$

Slack and Surplus Variables

Let the constraints of the general be

$$\sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, 2, \dots, k$$

Then the non-negative variables x_{n+i} , which satisfy

$$\sum_{j=1}^n a_{ij} x_j + x_{n+i} = b_i \quad i = 1, 2, \dots, k$$

are called slack variables.

Let the constraints of general be

$$\sum_{j=1}^n a_{ij} x_j \geq b_i, \quad i = k+1, k+2, \dots, l$$

Then the non-negative variables x_{n+i} which satisfies

$$\sum_{j=1}^n a_{ij} x_j - x_{n+i} = b_i \quad i = k+1, k+2, \dots, l$$

are called surplus variables.

Canonical and Standard forms of L.P.P

Canonical Form

The General LPP of the form given by

Maximize $z = a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n$

subject to the constraints.

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n \leq b_i, \quad i = 1, 2, \dots, m$$

$$x_j \geq 0, \quad j = 1, 2, \dots, n$$

is known as in canonical form

By using some elementary transformation, the general LPP can always put in the canonical form.

The characteristics of this form are:

Objective function is of the maximization type

The minimization of a function $f(x)$ is equivalent to the maximization of the negative expression of this function $-f(x)$

$$\text{ie Minimize } f(x) = \text{Maximize } \{ -f(x) \}$$

eg: The linear objective function

$$\text{Minimize } z = c_1x_1 + c_2x_2 + \dots + c_nx_n$$

is equivalent to

$$\text{Maximize } h = -z = -c_1x_1 - c_2x_2 - \dots - c_nx_n$$

$$\text{with } z = -h$$

All the constraints are of the type $\leq$, except for

the non-negative restrictions.

An inequality of $\geq$ type can be changed to an inequality of $\leq$ type by multiplying both sides of the inequality -1 .

For example, the linear constraint

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n \geq b_i$$

is equivalent to

$$-a_{i1}x_1 - a_{i2}x_2 - \dots - a_{in}x_n \leq -b_i$$

An equation may be replaced by two inequalities in opposite directions.

For example, the

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n = b_i \text{ is equivalent to}$$

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n \leq b_i$$

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n \geq b_i$$

3) All the variables are non-negative

A variable which is unrestricted in sign (i.e. +ve, -ve, 0) is equivalent to the difference between two non-negative variables.

Thus x_j is unrestricted in sign, it can be replaced by $(x_j' - x_j'')$ where x_j' and x_j'' are both non-negative i.e.

$$x_j = x_j' - x_j'', \text{ where } x_j' \geq 0, x_j'' \geq 0.$$

The standard Form

The General LPP of the form given by

$$\text{Maximize } z = c_1 x_1 + c_2 x_2 + \dots + c_n x_n$$

subject to constraints

$$a_{i1} x_1 + a_{i2} x_2 + \dots + a_{in} x_n = b_i \quad i = 1, 2, \dots, m$$

$$x_j \geq 0, \quad j = 1, 2, \dots, n$$

is known as in standard form.

The characteristics of this form are :-

1. Objective function, must be linear and either Maximize or Minimize. eg: $z = c_1 x_1 + c_2 x_2$.
2. All constraints must be equations with $\text{RHS} \geq 0$
use slack / surplus variables to convert $\leq$ and $\geq$
 $\leq \rightarrow \text{slack}$
 $\geq \rightarrow \text{surplus}$
3. All variables must be ≥ 0
eg: $x_1, x_2 \geq 0$.

Q. Reduce the following
standard form:

Maximize

$$x_0 = x_1 - 3x_2$$

Subject to the constraints

$$-x_1 + 2x_2 \leq 15$$

$$x_1 + 3x_2 = 10$$

x_1 and x_2 being unrestricted in sign

ans: Here x_1 and x_2 are unrestricted in sign

$$\text{Let } x_1 = x_1' - x_1'' \text{ and } x_2 = x_2' - x_2''$$

$$\text{where } x_1', x_1'', x_2', x_2'' \geq 0$$

Then the given LPP becomes

Maximize

$$x_0 = (x_1' - x_1'') - 3(x_2' - x_2'')$$

$$= x_1' - x_1'' - 3x_2' + 3x_2''$$

Subject to the constraints

$$-x_1' + x_1'' + 2x_2' - 2x_2'' \leq 15$$

$$x_1' - x_1'' + 3x_2' - 3x_2'' = 10$$

Introducing the slack variable $x_3 \geq 0$ in the
first inequality the standard form of given LPP

$$Ax = B$$

$$x \geq 0$$

$$Z = CX$$

$$Ax \leq B$$

$$x \geq 0$$

standard
form

canonical
form

is maximize.

$$z_{\text{max}} = x_1' - x_1'' - 3x_2' + 3x_2'' + 0 \cdot x_3$$

subject to the constraints:

$$-x_1' + x_1'' + 2x_2' - 2x_2'' + x_3 = 15$$

$$x_1' - x_1'' + 3x_2' - 3x_2'' = 10$$

where $x_1', x_1'', x_2', x_2'', x_3 \geq 0$

Q. Rewrite in standard form the following LPP.

Minimize

$$z = 12x_1 + 5x_2$$

subject to the constraints

$$5x_1 + 3x_2 \geq 15$$

$$7x_1 + 2x_2 \leq 10$$

$$x_1 \geq 0 \text{ and } x_2 \geq 0$$

ans. Here the objective function is of minimization type in the standard form of LPP objective function should be of maximization type. hence the objective function become

Maximize

$$-z = -12x_1 - 5x_2$$

$$h = -z = -12x_1 - 5x_2$$

Introducing surplus variable $x_3 \geq 0$ in the first constraint and slack variable $x_4 \geq 0$ in the second constraint the slot form of given LPP is given by.

Maximize

$$h = -12x_1 - 5x_2 + 0x_3 + 0x_4$$

subject to the constraints.

$$5x_1 + 3x_2 - x_3 = 15$$

$$7x_1 + 2x_2 + x_4 = 10$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0$$

Q. Reformulate the following LPP into the standard form.

$$\text{Maximize } z = 3x_1 + 4x_2 + 6x_3$$

subject to the constraints

$$2x_1 + x_2 + 2x_3 \geq 6$$

$$3x_1 + 2x_2 = 8$$

$$x_1 \geq 0, x_2 \geq 0$$

$$7x_1 - 3x_2 + 5x_3 \geq 9$$

and x_3 is unrestricted.

5. Here x_3 is unrestricted in sign.

$$\text{let } x_3 = x_3' - x_3''$$

$$\text{where } x_3', x_3'' \geq 0$$

then the LPP becomes

Maximize

$$z = 3x_1 + 4x_2 + 6x_3' - 6x_3''$$

Subject to the constraints

$$2x_1 + x_2 + 2x_3' - 2x_3'' \geq 6$$

$$3x_1 + 2x_2 = 8$$

$$x_1 \geq 0, x_2 \geq 0$$

$$7x_1 - 3x_2 + 5x_3' - 5x_3'' \geq 9$$

Introducing ^{1st} Surplus variable in ^{and 4th} First constraint

$$x_4 \geq 0 \text{ and } x_5 \geq 0$$

Then LPP is

Maximize

$$z = 3x_1 + 4x_2 + 6x_3' - 6x_3'' + 0 \cdot x_4 + 0 \cdot x_5$$

Subject to the constraints

$$2x_1 + x_2 + 2x_3' - 2x_3'' - x_4 = 6$$

$$3x_1 + 2x_2 = 8$$

$$7x_1 - 3x_2 + 5x_3' - 5x_3'' - x_5 = 9$$

$$x_1, x_2, x_3', x_3'', x_4, x_5 \geq 0.$$

Theorem

The set of Feasible solⁿ to a LPP is a convex set.

Proof:

Let the L.P.P be to determine x so as to Maximize the linear function.

$$\text{Maximize } z = cx, \quad c, x^T \in \mathbb{R}^n$$

subject to the constraints

$$Ax = b, \quad b \geq 0$$

$$x \geq 0$$

Let $x^{(1)}, x^{(2)}$ be two feasible solutions to this problem, so that

$$Ax^{(1)} = b$$

$$Ax^{(2)} = b$$

$$x^{(1)} \geq 0; x^{(2)} \geq 0$$

We will prove that convex combination of $x^{(1)}$ and $x^{(2)}$, namely

circle

Not

convex

Disk

convex



$$x = \lambda x^{(1)} + (1-\lambda) x^{(2)}, 0 \leq \lambda \leq 1$$

$\bar{x}$ is also feasible solution.

$$\begin{aligned} \text{consider } \Rightarrow Ax &= A(\lambda x^{(1)} + (1-\lambda) x^{(2)}) \\ &= \lambda Ax^{(1)} + (1-\lambda) Ax^{(2)} \\ &= \lambda b + (1-\lambda) b \\ &= \lambda b + b - \lambda b \\ &= b \end{aligned}$$

$$Ax = b$$

$$\text{And } x = \lambda x^{(1)} + (1-\lambda) x^{(2)} \geq 0 \quad [\lambda, (1-\lambda) \geq 0]$$

$$\therefore x \geq 0$$

Hence $\bar{x}$ is also a feasible solⁿ to the problem

Thus the set $S = \{x \mid x \text{ is a feasible solution to a L.P.P}\}$ is a convex set.

Basic Solution

Given a system of m simultaneously m linear equations in n unknowns ($m < n$)

$$Ax = b, \quad x^T \in \mathbb{R}^n$$

where A is an $m \times n$ matrix with rank m .

Let B be any $m \times m$ submatrix formed by m

linearly independent columns of A .

Then a solⁿ is obtained by setting $n-m$ variables, not associated with the columns, B equal to zero and solving the resulting system is called Basic solution to the given system of equations.

The m variables which may be all different from zero, are called Basic variables.

The $m \times m$ non singular ^{sub} matrix B is called a Basis matrix with column of B as Basis vectors.

Remarks

If B is the basis sub-matrix chosen, then the Basic solⁿ to the system is

$$x_B = B^{-1}b$$

But $x_B^T \in \mathbb{R}^m$, if x_B is a basic solⁿ of the given system the actual solⁿ is

$$[x_B^T, 0], 0 \in \mathbb{R}^{n-m}$$

If a Matrix with rank m then order m is non singular matrix

Q. Obtain all the basic solutions to the following systems of linear equations.

$$x_1 + 2x_2 + x_3 = 4$$

$$2x_1 + x_2 + 5x_3 = 5$$

ans: The matrix form of this linear equation is given by

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$\text{let } A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 5 \end{bmatrix}$$

not scalar multiple
 $\therefore$ linearly independent

$$\text{Here rank } A = 2$$

$$m = 2 \quad B_{m \times m} = B_{2 \times 2}$$

Take any of the following 2×2 submatrix as basis matrix B .

$$\begin{matrix} x_3=0 & x_1=0 & x_2=0 \end{matrix}$$
$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 5 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 2 & 5 \end{bmatrix}$$

The variable not associated with the columns of B are x_3 , x_1 and x_2 respectively

$$\text{Let us first take } B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

A basic solution to the given system is now

obtained by setting $x_3 = 0$.

And solving the system

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = B^{-1} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$B^{-1} = \frac{1}{|B|} \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{-3} \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$= \frac{-1}{3} \begin{bmatrix} -6 \\ -3 \end{bmatrix} = \underline{\underline{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}}$$

$$\therefore \underline{\underline{x_1 = 2 \quad x_2 = 1}}$$

Thus a basis solⁿ to the given system is

$x_1 = 2$, $x_2 = 1$ & $x_3 = 0$ (is non basic)

Let us take $B = \begin{bmatrix} 2 & 1 \\ 1 & 5 \end{bmatrix}$

$$x_1 = 0$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$B^{-1} = \frac{1}{|B|} \text{Adj}(B)$$

$$B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{Adj}(B) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 5 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 15 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 1.35 \\ 0.67 \end{bmatrix}$$

$$= \begin{bmatrix} 1.35 \\ 0.67 \end{bmatrix}$$

Thus a basis solⁿ to the given system is

$$x_2 = \frac{5}{9}, x_3 = \frac{2}{9} \text{ and non basic } x_1 = 0.$$

$$\text{let us take } B = \begin{bmatrix} 1 & 1 \\ 2 & 5 \end{bmatrix}$$

$$x_2 = 0$$

$$\begin{bmatrix} 1 & 1 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 5 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 15 \\ -3 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ -1 \end{bmatrix}$$

$$\begin{array}{r} 15x \\ 9 \\ \hline 135 \end{array}$$

Thus a basis solⁿ to the given system is
 $x_1 = 5$, $x_3 = -1$ and non basic $x_2 = 0$

Degenerate Solution

A basic solution to the system $AX=B$ is called Degenerate if one or more of the basic variables vanish.

Q. Show that the following system of linear equations has a degenerate solution.

$$2x_1 + x_2 - x_3 = 2$$

$$3x_1 + 2x_2 + x_3 = 3$$

ans. written in the matrix form as

$$AX = b.$$

where $A = \begin{bmatrix} 2 & 1 & -1 \\ 3 & 2 & 1 \end{bmatrix}$ $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ $b = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

Since $\text{rank } A = 2$.

The maximum no. of linearly independent columns of A is ~~here~~ 2.

Take any of the following 2×2 submatrix as

basis matrix B .

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}, \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix} \text{ and } \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$$

The variables not associated with the columns of B are x_3 , x_1 and x_2 respectively.

Let us first take $B = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$

A basic solution to the given system is now obtained by setting $x_3 = 0$ and solving the system

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$= \frac{1}{1} \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$= \underline{\underline{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}}$$

Thus a basic solⁿ to the given system is.

$$x_1 = 1, x_2 = 0.$$

=

$\therefore$ degenerate.

Let Take $\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$ here $x_1 = 0$

$$\begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$
$$= \frac{1}{3} \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{5}{3} \\ \frac{-1}{3} \end{bmatrix}$$

The basic solⁿ are $x_2 = 5/3$ $x_3 = -1/3$

Let Take $\begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$ here $x_2 = 0$

$$\begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$
$$= \frac{1}{5} \begin{bmatrix} 1 & 1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{5}{5} \\ 0 \end{bmatrix}$$

The basic solⁿ are

$$x_1 = 1$$

$$x_3 = 0$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

In each of the basic solution atleast one basic variable zero. Hence the basic solution are degenerate solution.

Basic Feasible solution

A feasible solution to a L.P.P which is also a basic solution to the problem is called basic feasible solution to the L.P.P.

Maximize the linear function

$$Z = CX, \quad C, x^T \in \mathbb{R}^n$$

Subject to the constraints

$$Ax = b$$

$$x \geq 0$$

x_B is a basic feasible solution to this problem

B is an $m \times m$ non-singular submatrix of A and B

$$x_B = b, \quad x_B \geq 0$$

Q. Obtain all the basic feasible solutions to the following system of linear equations.

a) $2x_1 + x_2 - x_3 = 2$

$3x_1 + 2x_2 + x_3 = 3.$

b.) $x_1 + 2x_2 + x_3 = 4$

$2x_1 + x_2 + 5x_3 = 5.$

ans: b) $x_1 + 2x_2 + x_3 = 4$

$2x_1 + x_2 + 5x_3 = 5$

The basic solⁿ of the given system are,

$x_1 = 2, x_2 = 1, x_3 = 0$ (non-basic) — ①

$x_2 = \frac{5}{3}, x_3 = \frac{2}{3}, x_1 = 0$ (" ") — ②

$x_1 = 5, x_3 = -1, x_2 = 0$ (" ") — ③

① and ② are basic feasible solution. but ③

is not a basic feasible solution. because $x_3 \neq 0$.

ans: a) $2x_1 + x_2 - x_3 = 2$

$3x_1 + 2x_2 + x_3 = 3.$

The basic solⁿ of the given system are,

$x_1 = 1, x_2 = 0, x_3 = 0$ ("non-basic") — ①

$x_2 = \frac{5}{3}, x_3 = -\frac{1}{3}, x_1 = 0$ (" ") — ②

$x_1 = 1, x_3 = 0, x_2 = 0$ (" ") — ③

① and ③ are basic feasible solution.
but ② is not a basic feasible solution
because $x_3 \neq 0$.

Associated cost vector

Let x_B be a basic feasible solution to the L.P.P.

Maximize

$$Z = CX$$

subject to the constraints

$$Ax = b, \quad b \geq 0$$

$$x \geq 0$$

Then the vector $C_B = (C_{B_1}, C_{B_2}, \dots, C_{B_m})$ where C_{B_i}

are components of C associated with

the basic variable is called the cost vector

associated with the basic feasible solution.

Remark:

The value of objective function for the basic feasible solution x_B is given by

$$Z_0 = C_B x_B$$

Improved Basic Feasible solution

Let x_B and $\hat{x}_B$ be two basic feasible solutions to the standard L.P.P then $\hat{x}_B$ is said to be an improved basic feasible solution as compared to x_B if $\hat{C}_B \hat{x}_B \geq C_B x_B$ where $\hat{C}_B$ constituted for cost components corresponding to $\hat{x}_B$.

Optimum basic feasible solution

A basic feasible solution x_B to the L.P.P.

Maximize

$$Z = Cx.$$

Subject to the constraints.

$$Ax = b, \quad b \geq 0$$

$$x \geq 0.$$

is called an optimum basic feasible solution if

$$Z_0 = C_B x_B \geq Z^*$$

where Z^* is value of the objective function for any feasible solution

Definition ;

Let x_B be a basic feasible solution to the L.P.P

Maximize

$$Z = CX$$

subject to the constraints

$$AX = b, b \geq 0$$

$$x \geq 0$$

Let C_B be the cost vector corresponding to x_B .

For each column vector a_j in A , which is not a

column vector of B , let

$$a_j = \sum_{i=1}^m y_{ij} b_i$$

Then the number

$$z_j = \sum_{i=1}^m C_B y_{ij}$$

is called the Evaluation corresponding to a_j .

and the number $z_j - C_j$ is called the net evaluation corresponding to a_j .

Fundamental properties of solutions

Theorem

The L.P.P has a feasible solution, then it also has a basic feasible solution.

Corollary

There exist only a finite no. of basic feasible solⁿ to a L.P.P.

Theorem

Let a L.P.P have a basic feasible solⁿ. If we drop one of the basis vector and introduce a non-basis vector in the basis set, then the new solution obtained is also a basic feasible solution.

Theorem (unbounded solution)

Let there exists a basic feasible solution to given L.P.P

If for atleast one j , for which $y_{ij} \leq 0$ ($i=1, 2, \dots, m$)

$Z_j - C_j$ is negative, then there does not exist any optimum solⁿ to this L.P.P.

Theorem

A necessary and sufficient condition for a basic feasible solution to a L.P.P to be an

optimum is that

$z_j - c_j \geq 0$ for all j for which the column vector a_j in A is not in the basis B .

Theorem

A basic feasible solution to a L.P.P must correspond to an extreme point of the set of all feasible solutions.

Theorem

Let F be a linear function of n variable then
minimum $f(x) = -\text{maximum}(-f(x))$

$$P = \{1, 2, 3\}$$

$$\min = -\max \{-1, -2, -3\}$$

$$\min = -\{-1\} = \underline{\underline{1}}$$

The Simplex Method

Q. Solve by Simplex Method

Maximize

$$Z = 7x_1 + 5x_2$$

Subject to constraints

$$x_1 + 2x_2 \leq 6$$

$$4x_1 + 3x_2 \leq 12$$

$$x_1, x_2 \geq 0$$

Step 1 $\rightarrow$ In the given LPP the objective function is of the maximization type.

Step 2 $\rightarrow$ In the given constraints the upper limits are non-negative.

Step 3 $\rightarrow$ The equation of the constraints can be converted into equations by introducing slack variables x_3 and x_4 respectively.

$$\text{i.e. } x_1 + 2x_2 + x_3 = 6$$

$$4x_1 + 3x_2 + x_4 = 12$$

Then the new objective function is

$$Z = 7x_1 + 5x_2 + 0x_3 + 0x_4$$

Step 4 $\rightarrow$ Find a basic feasible solⁿ. The set of equations can be written as.

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 4 & 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \end{bmatrix}$$

$$AX = B$$

clearly rank $A = 2$

Since $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ are two linearly independent

column vectors we can take $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is

a non-singular, non-negative basic submatrix of A .

Basic variables are x_3 and x_4 and a feasible solution

$$x_B = B^{-1}b.$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 12 \end{bmatrix}$$

$$= \begin{bmatrix} 6 \\ 12 \end{bmatrix}$$

$$x_B = \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \end{bmatrix}$$

E_B	y_B	x_B	y_1	y_2	y_3	y_4
0	y_3	$x_3 = 6$	1	2	1	0
0	y_4	$x_4 = 12$	(4)	3	0	1
coefficient $x_3 \& x_4$	$z_0 =$ $C_B x_B$	$z_0 = 0$	-7	-5	0	0

5) Compute the net evaluation $z_j - c_j$ ($c_j = 1, 2, \dots, n$)
by using the relation $z_j - c_j = C_B (y_j - y$
Examine the sign of $z_j - c_j$.

If all $z_j - c_j \geq 0$, bfs x_B is an optimum.

If atleast one < 0 , proceed to the next step.

step 6 $\rightarrow$ # If there are $z_0 = C_B X_B = [0 \ 0] \begin{bmatrix} 6 \\ 12 \end{bmatrix}$

More than 1 $z_j - c_j$ are -ve, we choose the most -ve of them (-7) which lies in the column y_1 . Since all the components of y_1 are +ve, the vector y_1 will enter the basis y_B (if no components are +ve the system has unbounded solutions).

$$\text{Compute } \frac{x_{OR}}{y_1} = \frac{6}{1} = 6 \quad \text{--- (1)}$$

$$\text{Choose the minimum of } \frac{12}{4} = 3 \quad \text{--- (2)}$$

$$\min(6, 3) = 3$$

$\therefore 4 \rightarrow$ leading element / pivotal element

Thus the vector y_4 leaves the basis.

4 is the leading element of the table.

C_B	y_B	x_B	y_1	y_2	y_3	y_4
0	y_3	$x_3 = 3$	0	$\frac{5}{4}$	1	$-\frac{1}{4}$
7	y_1	$x_1 = \frac{12}{4} = 3$	1	$\frac{3}{4}$	0	$\frac{1}{4}$
		$z_0 = 21$	0	$\frac{1}{4}$	0	$\frac{7}{4}$

Divide the 2nd row by 4 [leading element]

$$\hat{y}_{ij} = y_{ij} - \frac{y_{kj} \cdot y_{ir}}{y_{kr}}$$

$$x_3, x_3 = 6 - \left(\frac{12}{4} x_1\right) = 6 - 3 = \underline{3}$$

$$y_2, y_2 = 2 - \left(\frac{3}{4} x_1\right) = 2 - \frac{3}{4} = \underline{\frac{5}{4}}$$

$$y_3 = 1 - (0 x_1) = 1$$

$$y_4 = 0 - \left(\frac{1}{4} x_1\right) = \underline{-\frac{1}{4}}$$

$$y_1 = 1 - \left(\frac{4}{4} x_1\right) = \underline{0}$$

$Z_j - C_j$ are non-negative and hence an optimum solution has been reached. Thus an optimum basic

Feasible solution to the given LPP is,

$$x_1 = 3, x_2 = 0, x_3 = 3, x_4 = 0$$

Maximum

$$Z = 7x_1 + 5x_2 = 7 \times 3 + 5 \times 0 = 21$$

Q. Maximize

$$Z = 107x_1 + x_2 + 2x_3$$

Subject to the constraints,

$$14x_1 + x_2 - 6x_3 + 3x_4 = 7$$

$$16x_1 + \frac{1}{2}x_2 - 6x_3 \leq 5$$

$$3x_1 - x_2 - x_3 \leq 0$$

$$x_1, x_2, x_3, x_4 \geq 0$$

ans: Adding slack variables x_5 and x_6

$$\text{ie } 14x_1 + x_2 - 6x_3 + 3x_4 = 7$$

$$16x_1 + \frac{1}{2}x_2 - 6x_3 + x_5 = 5$$

$$3x_1 - x_2 - x_3 + x_6 = 0$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

$\therefore$ Maximize

$$Z = 107x_1 + x_2 + 2x_3 + 0x_4 + 0x_5 + 0x_6$$

Finding Basic feasible solution,

$$\begin{bmatrix} 14 & 1 & -6 & 3 & 0 & 0 \\ 16 & 1/2 & -6 & 0 & 1 & 0 \\ 3 & -1 & -1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \\ 0 \end{bmatrix}$$

Rank of $A = 3$

∴ choose a basis sub matrix of order 3.

To simplify the submatrix, multiply the 1st row by $1/3$.

$$\begin{bmatrix} 14/3 & 1/3 & -2 & 1 & 0 & 0 \\ 16 & 1/2 & -6 & 0 & 1 & 0 \\ 3 & -1 & -1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 7/3 \\ 5 \\ 0 \end{bmatrix}$$

$$\therefore B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x_B = B^{-1}b$$

$$\begin{bmatrix} x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 7/3 \\ 5 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 7/3 \\ 5 \\ 0 \end{bmatrix}$$

I

$$25, 5 - \frac{3}{0 \times 16} = 5$$

$$7/3 - 0 \times 1/3 = 7/3$$

$013 = 0$ $\hookrightarrow$ H_6 leaves the basis

$$5/16 = 0.3125$$

$$x_4 = \frac{14/3}{7/3} = \frac{1}{2} = 0.5$$

(3.2) $Q = \text{conversion}$

y_1 enters the basis
 $y_1 = 107$

[illegible]

$$J = Z_1^2 - C_1 = C_2 Y_2 - C_2$$

$$\underline{Z_0 = 0}$$

$$= [0, 0, 10, 7] \begin{bmatrix} 0 \\ 5 \\ 7/3 \end{bmatrix}$$

$$Z_0 = C_0 X_0$$

$$0 - (1/3) \times 16 = -16/3$$

$$1 - (0/3) \times 16 = 1$$

$$0 - (0/3) \times 16 = 0$$

$$-6 - (-1/3) \times 16 = -2/3$$

$$\frac{1}{2} - (-1/3) \times 16 = 35/6$$

2nd row

$$0 - (1/3) \times \frac{3}{4} = -\frac{1}{4}$$

$$0 - (0/3) \times \frac{3}{4} = 0$$

$$1 - (0/3) \times \frac{3}{4} = 1$$

$$-2 - (-1/3) \times \frac{3}{4} = -\frac{5}{4}$$

$$\frac{3}{4} - (-1/3) \times \frac{3}{4} = \frac{1}{4}$$

1st row

$$\frac{1}{3} + \frac{1}{4}$$

$$\frac{1}{3} + \frac{1}{4}$$

$$= [0, 0, 107] \begin{bmatrix} 17/9 \\ 35/6 \\ -1/3 \end{bmatrix} - 1 = \frac{-113}{3}$$

$$Z_3 - C_3 = C_B y_3 - C_3$$

$$= [0 \ 0 \ 107] \begin{bmatrix} -4/9 \\ -2/3 \\ -1/3 \end{bmatrix} - 2 = \frac{-113}{3}$$

$$Z_6 - C_6 = C_B y_6 - C_6$$

$$= [0 \ 0 \ 107] \begin{bmatrix} -14/9 \\ -16/3 \\ 1/3 \end{bmatrix} - 0 = \frac{107}{3}$$

$\frac{-113}{3}$ is the most negative corresponding column vector is y_3 but y_3 are all negative, $\therefore y_3$

will not enter the basis.

$\therefore$ This given LPP has an unbounded solution.

Q.

Maximize

$$Z = 4x_1 + x_2 + 4x_3 + 5x_4$$

subject to

$$4x_1 - 6x_2 - 5x_3 + 4x_4 \geq -20$$

$$3x_1 - 2x_2 + 4x_3 + x_4 \leq 10$$

$$8x_1 - 3x_2 + 3x_3 + 2x_4 \leq 20$$

$$x_1, x_2, x_3, x_4 \geq 0$$

ans: First constraint is -ve so multiply by -1

$$-4x_1 + 6x_2 + 5x_3 - 4x_4 \leq 20$$

Adding slack variable x_5, x_6, x_7 to the

given constraints. Then given LPP becomes: Maximize

$$-4x_1 + 6x_2 + 5x_3 - 4x_4 + x_5 = 20$$

$$3x_1 - 2x_2 + 4x_3 + x_4 + x_6 = 10$$

$$8x_1 - 3x_2 + 3x_3 + 2x_4 + x_7 = 20$$

Maximize

$$Z = 4x_1 + x_2 + 4x_3 + 5x_4 + 0x_5 + 0x_6 + 0x_7$$

Find the basic feasible solⁿ.

The matrix form given LPP.

$$AX = b$$

$$\begin{bmatrix} -4 & 6 & 5 & -4 & 1 & 0 & 0 \\ 3 & -2 & 4 & 1 & 0 & 1 & 0 \\ 8 & -3 & 3 & 2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \\ 20 \end{bmatrix}$$

Rank of $A = 3$

Take 3×3 non singular submatrix B is

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$x_B = B^{-1}b$, Basic Feasible solⁿ.

$$\begin{bmatrix} x_5 \\ x_6 \\ x_7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 20 \\ 10 \\ 20 \end{bmatrix}$$

$$\begin{bmatrix} x_5 \\ x_6 \\ x_7 \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \\ 20 \end{bmatrix}$$

① $x_j - c_j$

② x_5, x_6, x_7
= 0

③ $x_j - c_j =$
 $c_B y_B - c_j$

Simplex Table c_j

	c_1	c_2	c_3	c_4	c_5	c_6	c_7
	4	1	4	5	0	0	0

c_B	y_B	x_B	y_1	y_2	y_3	y_4	y_5	y_6	y_7
0	y_5	$x_5 = 20$	-4	6	5	-4	1	0	0
0	y_6	$x_6 = 10$	3	-2	4	1	0	1	0
0	y_7	$x_7 = 20$	8	-3	3	2	0	0	1
$Z = 0$			-4	-1	-4	-5	0	0	0

Most -ve $z_j - c_j = -5$, corresponding column vector

y_4 will enter and 2 of the

$$\text{Min} = \left\{ \frac{10}{1}, \frac{20}{2} \right\} = \{10, 10\}$$

choose $y_{24} = 1$ as the leading element.

y_6 leaves the basis y_4 enters.

			4	1	4	5	0	0	0
c_B	y_B	x_B	y_1	y_2	y_3	y_4	y_5	y_6	y_7
0	y_5	$x_5 = 60$	3	-2	21	0	1	4	0
5	y_4	$x_4 = 10$	3	-2	4	1	0	1	0
0	y_7	$x_7 = 0$	2	<u>1</u>	-5	0	0	-2	1
		z_0	11	-11	16	0	0	5	0

$z_j - c_j$

Most -ve = x_2

$$z_{y_1} = 15$$

so another iteration is needed.

$$z_{y_2} = 10$$

Ratio

$$z_{y_3} = 20$$

$$z_{y_4} = 5$$

$$z_{y_5} = 0$$

$$z_{y_6} = 5$$

$$z_{y_7} = 0$$

$$-2/60/2 = -1/30 \quad -60/2 = -30$$

$$-10/10/2 = -1/5 \quad -10/2 = -5$$

$$1/0 = \infty \quad 0/1 = 0$$

$$-4 - \left(\frac{3}{1}\right) = -4$$

$$= -4 + 12 = 8$$

$$6 - \left(\frac{-2}{1}\right) = 4$$

$$-3 - (2 \times -2)$$

$$-3 + 4 = 1$$

y_7 leaves y_2 enters

CB	y_0	x_B	y_1	y_2	y_3	y_4	y_5	y_6	y_7
0	y_5	$x_5 = 60$	12	0	<u>11</u>	0	1	0	2
5	y_4	$x_4 = 10$	7	0	-6	1	0	-3	2
1	y_2	$x_2 = 0$	2	1	-5	0	0	-2	1
		Z_0	33	0	<u>-39</u>	0	0	-17	11

$z_j - c_j$

Most -ve = -39

$z_j = c_B \times x_B$

Ratio

$y_1 = 37$

$60/11 = 60/11 \checkmark$

$y_2 = 1$

$10/-6 = -10/6$ not allowed.

$y_3 = -35$

$0/-5 = 0$ not allowed

$y_4 = 5$

$y_5 = 0$

y_3 enters and y_5 leaves

$y_6 = -17$

$y_7 = 11$

-ve

denominator

not allowed

CB	y_0	x_B	y_1	y_2	y_3	y_4	y_5	y_6	y_7
4	y_3	$x_3 = \frac{60}{11}$	$12/11$	0	1	0	$1/11$	0	$2/11$
5	y_4	$x_4 = \frac{470}{11}$	$149/11$	0	0	1	$6/11$	-3	$34/11$
1	y_2	$x_2 = \frac{300}{11}$	$82/11$	1	0	0	$3/11$	-2	$21/11$
		Z_0	$831/11$	1	4	5	$39/11$	-17	$199/11$

$y_1 = 4 - \left[\frac{48}{11} + \frac{149 \times 5}{11} + \frac{82}{11} \right] = \frac{-831}{11}$

Most -ve -17

y_6 enters

Q. Maximize $z = 4x_1 + 10x_2$

Subject to the constraints

$$2x_1 + x_2 \leq 50$$

$$2x_1 + 5x_2 \leq 100$$

$$2x_1 + 3x_2 \leq 90$$

$$x_1 \geq 0, x_2 \geq 0$$

ans: Add slack variables:

$$2x_1 + x_2 + x_3 = 50$$

$$2x_1 + 5x_2 + x_4 = 100$$

$$2x_1 + 3x_2 + x_5 = 90$$

Objective function

$$z = 4x_1 + 10x_2 + 0x_3 + 0x_4 + 0x_5$$

Find the basic feasible solⁿ.

The matrix for of given LPP.

$$AX = b.$$

$$\begin{bmatrix} 2 & 1 & 1 & 0 & 0 \\ 2 & 5 & 0 & 1 & 0 \\ 2 & 3 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 50 \\ 100 \\ 90 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$XB = B^{-1}b.$$

$$\begin{bmatrix} x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 50 \\ 100 \\ 90 \end{bmatrix}$$

$$\begin{bmatrix} x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 50 \\ 100 \\ 90 \end{bmatrix}$$

$C_j = 4 \quad 10 \quad 0 \quad 0 \quad 0$							
C_B	Y_B	X_B	y_1	y_2	y_3	y_4	y_5
0	y_3	$x_3=50$	2	1	1	0	0
0	y_4	$x_4=100$	2	5	0	1	0
0	y_5	$x_5=90$	2	3	0	0	1
		$Z_0=0$	-4	-10	0	0	0

Most -ve = -10

Ratio

$$\frac{50}{1} = 50$$

$$\frac{100}{5} = 20$$

$$\frac{90}{3} = 30$$

$$\text{Min} = \{50, 20, 30\} = 20.$$

∴ y_2 enters and y_4 leaves.

			C_j	4	10	0	0	0
CB	C_B	X_B	y_1	y_2	y_3	y_4	y_5	
0	43	$x_3 = 30$	$8/5$	0	1	$-1/5$	0	
10	42	$x_2 = 20$	$2/5$	1	0	$1/5$	0	
0	45	$x_5 = 30$	$4/5$	0	0	$-3/5$	1	
		$Z_j - C_j$	0	0	0	+2	0	

There is no negative value in $Z_j - C_j$

∴ optimal solution reached

$$x_1 = 0, x_2 = 20, x_3 = 30, x_4 = 0, x_5 = 30$$

Maximum value .

$$Z = 4x_1 + 10x_2 = 4(0) + 10(20) = 200$$

$$x_1 = 0, x_2 = 20$$

$$Z_{\max} = \underline{\underline{200}}$$